Lecture 55 : SEI CMM (Contd.)

Introduction

In the dynamic world of software development, ensuring high quality is paramount for project success and organizational reputation. Over recent lectures, we've explored the critical role of robust quality management systems. An organization's commitment to quality is often reflected in its adherence to established quality standards. While ISO 9001 provides a broad framework for quality management across various industries, the Software Engineering Institute's Capability Maturity Model (SEI CMM) offers a specialized, step-by-step approach tailored specifically for software development. This lecture delves deeper into the SEI CMM, focusing on its Key Process Areas (KPAs) which guide organizations through a structured journey of quality improvement across its five maturity levels. Understanding these KPAs is crucial for any organization aiming to systematically enhance its software development processes, moving from an ad-hoc, chaotic state to a mature, optimized, and continuously improving environment. We will also compare SEI CMM with ISO 9001 and explore its evolution into CMMI, providing a comprehensive view of industry-recognized quality benchmarks.

Core Concepts

Term	Definition	Significance
`SEI CMM`	A framework developed by the Software Engineering Institute that describes five evolutionary levels of software process maturity, from ad-hoc to optimized.	Provides a structured path for organizations to improve their software development and maintenance processes systematically.
`Maturity Levels`	The five distinct stages (Initial, Repeatable, Defined, Managed, Optimizing) in the SEI CMM, representing increasing levels of process capability and control.	Each level signifies a progressive improvement in an organization's ability to manage and develop software, with higher levels indicating greater predictability and quality.
	A cluster of related activities that, when performed	KPAs define specific areas an organization must focus on to

Term	Definition	Significance
`Key Process Area (KPA)`	collectively, achieve a set of goals considered essential for improving process capability at a particular maturity level.	advance from one maturity level to the next, providing a roadmap for improvement.
`ISO 9001`	An international standard for quality management systems, applicable across all industries, focusing on meeting customer and regulatory requirements.	Offers a generic quality assurance framework, but is less specific to software development challenges compared to SEI CMM.
`CMMI`	Capability Maturity Model Integration, an integrated framework that combines multiple CMMs into a single model, addressing development, services, and acquisition.	Aims to resolve the confusion and complexity arising from multiple, disparate CMMs by providing a unified, comprehensive approach to process improvement.
`Quantitative Process Management`	A practice at higher maturity levels where process and product metrics are systematically collected, analyzed, and used to manage and control processes.	Enables data-driven decision-making, allowing organizations to identify and correct causes of variation and predict process performance.
`Defect Prevention`	A Level 5 KPA focused on identifying the root causes of defects and proactively changing processes to prevent their recurrence.	Represents a shift from defect detection to defect avoidance, leading to significant improvements in product quality and efficiency.

Detailed Analysis

1. Key Process Areas (KPAs) by Maturity Level

The SEI CMM defines specific Key Process Areas (KPAs) that an organization must master to progress from one maturity level to the next. These KPAs provide a roadmap for systematic quality improvement.

- **Level 1: Initial**
- There are no defined KPAs for Level 1. This is because every organization, by default, starts at Level 1, characterized by an ad-hoc, often chaotic, process where success depends on individual heroics rather than established procedures. No specific process areas are identified for improvement at this foundational stage, as the very concept of a defined process is absent.
- **Level 2: Repeatable**
- To move from Level 1 to Level 2, an organization must establish basic project management controls, ensuring that successful practices can be repeated. The KPAs for Level 2 are fundamental to any well-managed software project:
- **Requirements Management**: This KPA emphasizes the need for proper requirement specification and controlled management of changes to these requirements. Requirements form the basis for all planning and management activities in a software project.
- **Software Project Planning**: Focuses on establishing realistic plans for software projects, including estimating size, cost, and developing schedules. This is a core aspect of software project management.
- **Software Project Tracking and Oversight**: Involves monitoring the project's progress against the established plans and taking corrective actions when deviations occur. This ensures that projects stay on track.
- **Software Subcontract Management**: Addresses the process of managing work outsourced to external vendors. It ensures that contracts are clearly specified and deliverables are checked against specifications.
- **Software Quality Assurance (SQA)**: Establishes an independent function to objectively review and audit software products and activities to ensure compliance with established procedures.
- **Software Configuration Management (SCM)**: Requires the use of tools and procedures to control changes to software artifacts (e.g., requirements, design, code). This ensures that authorized changes are tracked, and the latest versions are identifiable.

- **Level 3: Defined**
- Organizations at Level 3 move beyond project-specific practices to establish a standardized, documented software process for the entire organization. The KPAs include:
- **Organization Process Focus**: Establishes organizational responsibility for software process activities that improve the organization's overall software process capability.
- **Organization Process Definition**: Develops and maintains a standard set of software processes for the organization, often compiled into a "process handbook." This ensures consistency across projects.
- **Training Program**: Recognizes that software development skills are constantly evolving. This KPA emphasizes providing periodic training to developers to keep them updated with the latest technologies and methods.
- **Integrated Software Management**: Tailors the organization's standard software process to specific projects and manages the project according to the tailored process.
- **Software Product Engineering**: Defines and integrates the technical activities of software engineering, including requirements analysis, design, code, and test.
- **Intergroup Coordination**: Establishes a means for the software engineering group to participate with other project groups to address project-wide issues.
- **Peer Reviews**: Encourages systematic reviews of work products (e.g., requirements, design, code) by peers to detect and remove defects early in the development lifecycle.
- **Level 4: Managed**
- At Level 4, organizations achieve quantitative control over their software processes and products. This involves collecting and analyzing data to understand and predict process performance.
- **Quantitative Process Management**: Focuses on controlling the process performance quantitatively. This involves collecting process and product metrics to identify variations and their causes. For example, if the defect rate after testing is high, the project manager investigates why, perhaps due to insufficient testing rigor.
- **Software Quality Management**: Establishes a quantitative understanding of the quality of the software product and achieves specific quality goals.
- Examples of metrics used include:

- Estimated vs. actual lines of code: Significant variations indicate issues in estimation or development.
- Number of issues raised during code inspection: A low number might suggest ineffective inspections if many defects are found later.
- Number of defects detected during unit testing: High numbers here might indicate poor coding practices.
- Number of defects detected during system testing: High numbers here could point to failures in earlier testing phases or design flaws.
- The focus here is on identifying and correcting causes of variation within the existing process, not necessarily changing the process itself.
- **Level 5: Optimizing**
- Level 5 organizations are focused on continuous process improvement, proactively preventing defects, and incorporating new technologies.
- **Defect Prevention**: This KPA goes beyond identifying variations to analyzing the root causes of defects and making fundamental changes to the process to prevent their recurrence. It's about improving the process itself, not just managing its current performance.
- **Technology Change Management**: Recognizes the rapid evolution of software technology. This KPA ensures that the organization actively identifies, evaluates, and incorporates beneficial new technologies, tools, and methods into its processes.
- **Process Change Management**: Since processes are continually evolving in a Level 5 organization, this KPA ensures that the process itself is managed under configuration control. This allows for tracking different versions of processes, ensuring clarity on the latest or applicable process for various teams.

2. Comparison of SEI CMM and ISO 9001

While both SEI CMM and ISO 9001 aim to improve quality, they differ significantly in scope, application, and approach:

- **Certification vs. Assessment**: ISO 9001 is a certification awarded by international standards bodies, allowing organizations to officially quote their compliance. SEI CMM, on the other hand, is an assessment primarily for internal use, guiding an organization's self-improvement or for a vendor selection process.
- **Industry Focus**: ISO 9001 is a generic standard applicable across all industries (manufacturing, services, etc.). SEI CMM is specifically tailored for

the software industry, addressing unique challenges and nuances of software development.

- **Quality Focus**: The original ISO 9001 (1987 version) primarily targeted quality assurance, roughly corresponding to Level 3 of SEI CMM. Later versions (e.g., ISO 9001:2000) expanded to target Total Quality Management (TQM). SEI CMM, from its inception, aimed for continuous improvement towards TQM.
- **Improvement Approach**: ISO 9001 offers a "go/no-go" solution; an organization either qualifies or does not. SEI CMM provides a step-by-step, gradual approach through its five maturity levels, offering a clear path for continuous quality improvement. It emphasizes that skipping levels or KPAs can be counterproductive, as foundational practices must be in place before advanced ones.
- **Third-Party vs. Internal**: ISO 9001 certification is typically granted by accredited third-party agencies. SEI CMM assessments are often conducted internally or by authorized lead assessors, without a formal "certification" in the same sense as ISO.

3. Evolution to CMMI

The proliferation of various CMMs (e.g., CMM for software development, CMM for systems engineering, People CMM, CMM for software acquisition) led to confusion. Organizations dealing with multiple domains found it challenging to apply different models with varying terminologies and structures. To address this, the Capability Maturity Model Integration (CMMI) was proposed in 2002 and has been revised multiple times since.

- **Integration of Models**: CMMI's main idea was to integrate a large number of disparate CMMs into a single, unified framework. This provided a consistent approach to process improvement across different organizational functions.
- **Three Flavors of CMMI**: CMMI evolved into different "flavors" to cater to specific organizational needs:
- **CMMI for Development (CMMI-DEV)**: Applicable to organizations developing products and services, encompassing software engineering, systems engineering, and collaborative teams.
- **CMMI for Services (CMMI-SVC)**: Designed for organizations providing services, ranging from call centers to taxi aggregators.
- **CMMI for Acquisition (CMMI-ACQ)**: Used by organizations that acquire products and services from suppliers.

This integration aimed to simplify process improvement efforts, making it easier for organizations to apply maturity models consistently across their diverse operations.

Key Takeaways

- [ ] Effective software quality management requires a structured quality system, with SEI CMM offering a specialized framework for software.
- [ ] SEI CMM defines five maturity levels, each representing increasing process capability, from ad-hoc (Level 1) to continuously optimizing (Level 5).
- [ ] Key Process Areas (KPAs) are specific sets of activities that an organization must implement to advance from one maturity level to the next.
- [ ] Level 2 KPAs focus on basic project management, including requirements, planning, tracking, configuration management, and subcontract management.
- [ ] Level 3 KPAs emphasize organizational-wide process definition, documentation, training, and peer reviews to standardize practices.
- [ ] Level 4 KPAs introduce quantitative process management, using metrics to measure and control process performance and identify variations.
- [ ] Level 5 KPAs focus on continuous improvement through defect prevention, technology change management, and managing process changes itself.
- [ ] SEI CMM provides a step-by-step improvement path, unlike ISO 9001's go/no-go certification, and is software-specific.
- [ ] The evolution to CMMI aimed to integrate multiple CMMs into a single, unified framework to reduce complexity and provide consistency across different organizational functions (development, services, acquisition).

Conclusion

The SEI CMM, and its successor CMMI, provide invaluable frameworks for organizations committed to achieving excellence in software development and service delivery. By systematically addressing the Key Process Areas at each maturity level, organizations can transform their development practices from unpredictable and reactive to well-defined, quantitatively managed, and continuously improving. This structured approach not only enhances product quality and predictability but also fosters a culture of continuous learning and adaptation to technological advancements. While ISO 9001 offers a broad quality assurance standard, the CMM models provide the granular, software-specific guidance

necessary for navigating the complexities of modern software engineering. As we conclude our discussion on SEI CMM, we recognize that these models lay the groundwork for individual practitioners to also adopt similar disciplined approaches, which will be the focus of our next lecture on the Personal Software Process (PSP).